



Fig. 8: Component layout for the PCB in Fig. 7

Power supply. Fig. 3 shows the power supply circuit. The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 1A. The transformer output is rectified by a full-wave rectifier comprising diodes D8 through D11, filtered by capacitor C6 and regulated by IC 7808 (IC18). A 6V rechargeable battery is used for battery backup. The battery is charged through diode D14. Diode D15 is reverse-biased when the regulator output is present. The battery powers the circuit through diode D15 during no regulator output.

Construction

An actual-size, single-side PCB of analogue clock with add-on calendar, except circuit-inside the dotted line (circuit around IC14 and IC15), is shown in Fig. 5 and its component layout in Fig. 6. The circuit built around IC14 and IC15 (inside the dotted line) is used for month, year and date displays. Separate PCB should be used for date, month and year displays. An actual-size, single-side PCB for the circuit around IC14 and IC15 (inside the dotted line) is shown in Fig. 7 and its component layout in Fig. 8.

You can assemble the complete circuit on a single PCB or build individual circuit blocks, mount them as per the space available and wire together. It is easy to build individual circuit blocks and check them for proper functioning, by connecting them one after the other. It is also easier to troubleshoot any problem. The only disadvantage is that you may have to run more wires to connect them.

Use a suitable box for the clock. Mount on it the displays, setting switches, etc. (A readily available 30.5x25.4x6.4cm³ electrical switch box with white bakelite board top was used by the author.) Fit the transformer and backup battery inside the box.

De-bouncing circuits have been used at various toggle switches to eliminate the contact bouncing effect. All the clock input points should be connected to ground through the combination of a 100-kilo-ohm resistor and a 0.01µF capacitor to ground to prevent any unwanted spike. Further, use a 100µF capacitor along with a 0.1µF capacitor at supply points of individual circuit blocks.

Use proper heat-sink for the voltage regulator IC. Fix the transformer away from the clock generator circuit in order to avoid 50Hz pickup. Also avoid any wiring around the transformer. Solder the components neatly with flux within the fine tracks of the PCB. Use extra length of wires for wiring the switches, to

avoid pulling them while opening/fixing the circuits. Check the circuit for any shorting of wires before connecting the power supply to it.

Setting the clock

Set the time in the following order: Minutes→Hours→Date→Day. Before setting these functions, you need to reset the respective counters. Month and year can be set independently.

Select the required function with the help of the respective toggle switch. Once the function is set, put the switch back to its normal position, to run the clock in normal mode. The setting may advance by one count when the toggle switch is changed to normal position. If this happens, repeat the setting and accordingly set one count less and change the toggle switch to normal position to get the required setting. If the date and day are not synchronised by default, change synchronisation switch S14 to the other position. ●

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